Moed b solution

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1.1

Two-dimensional wave-equation:

$$\nabla^2\psi\left(x,y,t\right)+\frac{1}{c^2}\ddot{\psi}=0$$

$$\nabla^2\equiv\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}$$

$$\omega^2=c\left|\vec{k}^2\right|\quad\vec{k}=k_x\hat{x}+k_y\hat{y}$$

$$c=\sqrt{\frac{\omega^2}{k_x^2+k_y^2}}$$

Basis functions:

$$u\left(x,y;k_x,k_y\right)=\frac{\exp\left(i\vec{k}\cdot\vec{r}\right)}{2\pi}=\frac{\exp\left(i\left[k_xx+k_yy\right]\right)}{2\pi}\quad\vec{r}=\begin{pmatrix}x\\y\end{pmatrix}$$

Units:

$$\left[\psi\right]=m$$

$$\left[k\right]=m^{-1}$$

$$\left[\omega\right]=s^{-1}$$

$$\left[c\right]=m\,s^{-1}$$

1.2

Incident wave:

$$\psi_i=f\left(x,y,t\right)=f\left(x-c_xt,y-c_yt\right)$$

$$\hat{k}_i=\hat{x}+\hat{y}$$

Reflected wave:

$$\psi_r\left(x,y,t\right)=-f\left(x+c_xt,y-c_yt\right)$$

$$\hat{k}_r = -\hat{x} + \hat{y}$$

Reflection coefficient:

$$R = -1$$

1.3

$$\omega(k) = \alpha \frac{c}{1+\epsilon k}$$

$$[\alpha] = m^{-1}$$

$$[\epsilon] = m$$

$$\boxed{\beta = \frac{\partial^2 \omega}{\partial k^2} = \frac{2\alpha\epsilon^2 c}{(1+\epsilon k)^3}}$$

$$\boxed{[\beta] = m^2 s^{-1}}$$

Dimensions check:

$$\left[\frac{2\alpha\epsilon^2 c}{(1+\epsilon k)^3}\right] = \cancel{m^{-1}} m^2 \cancel{m} s^{-1} = m^2 s^{-1}$$

Width:

$$\psi(\tilde{x}, t) \propto e^{i(k_0 x - \omega t)} \exp\left(-\frac{1}{2}\frac{(x - v_g t)^2}{\sigma^2 + 2i\beta t}\right) + c.c.$$

$$\propto \exp\left(-\frac{1}{2}\frac{(x - v_g t)^2}{\sigma^4 + 4\beta^2 t^2}\left(\sigma^2 - 2i\beta t\right)\right) + c.c.$$

$$= \exp\left(-\frac{1}{2}\frac{(x - v_g t)^2}{\sigma^4 + 4\beta^2 t^2}\left(\sigma^2 - 2i\beta t\right)\right) + c.c. = \exp\left(-\frac{1}{2}\frac{(x - v_g t)^2}{\sigma^2 + 4\frac{\beta^2 t^2}{\sigma^2}}\right) e^{i\phi(t)} + c.c.$$

$$\boxed{\sigma(t) = \sqrt{\sigma^2 + 4\frac{\beta^2 t^2}{\sigma^2}}}$$

1.4

Direction\polarization: $\hat{z}$

If we add elasticity, then the polarization can be in all three directions, forming the following basis: $\hat{x}, \hat{y}, \hat{z}$

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2.1

$$\tau = \frac{x_0}{c}$$

For short times $t \ll \tau$ we solve using d'Alembert's Solution.

$$u(x,t) = \frac{1}{2}\left(u_0(x-ct) + u_0(x+ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(z)\,dz$$

$$\Downarrow$$

$$u(x,t) = \frac{A}{2}\exp\left(-\frac{1}{2}\frac{(x-x_0-ct)^2}{\sigma^2}\right)\sin\left(k_0(x-ct)\right)$$

$$+\frac{A}{2}\exp\left(-\frac{1}{2}\frac{(x-x_0+ct)^2}{\sigma^2}\right)\sin\left(k_0(x+ct)\right)$$

2.2

2.3

$$u(x,t) = R\frac{A}{2}\exp\left(-\frac{1}{2}\frac{(x-c(t-\tau))^2}{\sigma^2}\right)\sin\left(k_0(x+x_0-ct)\right)$$

$$+T\frac{A}{4\pi}\frac{\sigma}{\sqrt{\sigma^2+i2\beta t}}\exp\left(-\frac{1}{2}\frac{(x+v_g(t-\tau))^2}{\sigma^2+i2\beta t}\right)e^{i\tilde{k}_0(x+v_p(t-\tau))} + c.c.$$

$$\omega(k_0) = \alpha k_0^2$$

$$ck_0 = \alpha\tilde{k}_0^2 \Rightarrow \tilde{k}_0 = \sqrt{\frac{ck_0}{\alpha}}$$

$$v_p = 2\alpha\tilde{k}_0 = 2\sqrt{\alpha c k_0}$$

$$\beta = 2\alpha$$